Technical Documentation

Product	**InFocus IN1100,IN1102,M20,M22,LP85,LP85W,M11,M11W**
Title	**IR Codes & Documentation**

Introduction:

IR (infrared) light can be used to send commands to and retrieve status information from many consumer devices. IR communication can be used for two-way data communication (mostly between computers and PDAs), though most display devices, including projectors, are configured to receive commands only. Commonly the IR capability of projectors is used in conjunction with the factory-provided remote control. However, if there are multiple devices that need control, you may wish to use a universal or learning remote.

Many universal remote controls can only control devices that the remote manufacturer has included in their table of pre-programmed codes. These types of universal remote controls aren't usually capable of controlling projectors and other less common equipment. A universal remote with a "Learning" function is suitable for use with projectors and other unique components (game consoles, media PCs & extenders, curtain/lighting controls, etc.). Many other programmable remotes are also able to work with projectors and other equipment.

If you need help setting up your universal "Learning" remote, please refer to the universal remote manufacturer's documentation.

The directions below are intended to help you set up a programmable remote, using the codes for your InFocus projector.

Configuration:

- InFocus projectors use the NEC IR protocol.
- The customer code portion of the IR packet should use the hexadecimal value: 0x4E87
- Please note that the letter O is not used in hexadecimal, only zeros are used.

Supported Commands:

The following table lists all of the currently designated infrared key codes.

Key Name	Description	Hexadecimal
Menu	Toggles menu state up or down	02h
Brightness Plus	Increases brightness	03h
Volume Minus	Decreases volume	04h
Brightness Minus	Decreases brightness	07h
Volume Plus	Increases volume	08h
Focus Near	Commands the Motorized Zoom to Focus Near	09h
Focus Far	Commands the Motorized Zoom to Focus Far	0Ah
Zoom In	Commands the Motorized Zoom to Zoom In	0Bh
Zoom Out	Commands the Motorized Zoom to Zoom Out	0Ch
Standby	Toggles standby state on or off	0Dh
Freeze	Toggles freeze state on or off	0Eh
Light	Enables backlighting on remote control **(not transmitted)**	0Fh
Source Select	Selects next available source	10h
Cursor Left	Moves menu cursor left	11h
Cursor Right	Moves menu cursor right	12h
Cursor Up	Moves menu cursor up	13h
Cursor Down	Moves menu cursor down	14h
Power	Toggles power state on or off	17h
Reset Bulb Counter	Resets lamp hour counter (**not transmitted**)	18h
Computer Toggle	Selects next available computer source	19h
Video Toggle	Selects next available video source	1Ah
Horizontal Offset Left	Commands the Motorized Zoom to move left	1Bh
Horizontal Offset Right	Commands the Motorized Zoom to Move Right	1Ch
Vertical Offset Up	Commands the Motorized Zoom to Move Up	1Dh
Vertical Offset Down	Commands the Motorized Zoom to Move Down	1Eh
Keystone Plus	Increases keystone	20h
Keystone Minus	Decreases keystone	21h
Next Slide	Moves to next slide in powerpoint (**sends PageUp via USB**)	24h
Previous Slide	Moves to previous slide in powerpoint (**sends PageDn via USB**)	25h
Zoom Plus	Increases zoom ratio	26h
Zoom Minus	Decreases zoom ratio	27h
Zoom Toggle	Toggles zoom state on or off	28h
New Mute	Mutes audio	29h
USB Flash	Enables USB Flash mode (**not transmitted**)	2Ah
Effect	Activates programmed effect	2Bh
Presets	Toggles through available presets	2Ch
Source1	Selects programmed source	2Dh
Source2	Selects programmed source	2Eh
Source3	Selects programmed source	2Fh
Source4	Selects programmed source	30h
Blank	Toggles state of blank screen on or off	31h
Select	Selects current menu item	32h
Auto Image	Activates automatic image re-acquisition	33h
Resize	Toggles through available resize modes	34h

Overscan	Toggles overscan state on or off	35h
Contrast Plus	Increases contrast	36h
Contrast Minus	Decreases contrast	37h
Discrete Power Off	Commands the system to power down	39h
Discrete Power On	Commands the system to power up	3Ah
Discrete Resize Native	Selects the Native aspect ratio	3Bh
Discrete Resize 16:9	Selects the 16:9 aspect ratio	3Ch
Discrete Resize 4:3	Selects the 4:3 aspect ratio	3Dh
Discrete Resize Letterbox	Selects the Letterbox aspect ratio	3Eh
Discrete Resize Natural Wide	Selects the Natural Wide aspect ratio	3Fh
Discrete User Preset 1	Selects the User Preset 1	40h
Discrete User Preset 2	Selects the User Preset 2	41h
Discrete User Preset 3	Selects the User Preset 3	42h
Discrete Blank Off	Turns off the blank screen	43h
Discrete Blank On	Turns on the blank screen	44h